

PAPER Sub-Millisecond Neural Equilibrium Reconstruction Robust to Sensor Dropout: A Virtual-Sensor Mesh for Radiation-Degraded Diagnostics in a Compact Spherical-Tokamak Breeder

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Abstract

Every control decision in a compact spherical tokamak begins with the equilibrium, and in a nuclear device that equilibrium must be reconstructed faster than the plant, verified against the governing physics, and kept trustworthy when a diagnostic degrades or drops out under a neutron field. We formalise and demonstrate a real-time reconstruction stack for the Hyperion breeder (design point $I_p = 9.66$ MA, $Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, negative triangularity $\delta = -0.30$, $B_0 = 8$ T, elongation $\kappa = 2.0$, major radius $R_0 = 1.20$ m, aspect ratio $A = 2.5$, centrepost peak field 16.84 T). A physics-informed neural surrogate reconstructs the poloidal flux $\psi(R, Z)$ to a root-mean-square error of **0.139%** of range against a free-boundary Grad–Shafranov reference — inside the 1% acceptance target and improving a prior 1.79% attempt — at a **2877×** speedup (0.87 ms inference versus a ~ 2.5 s Picard solve), with a discretised Grad–Shafranov operator residual of 0.32% on the interior confirming the reconstruction is a solution of the governing equation and not a free-form fit. A graph-neural virtual-sensor mesh assimilates the diagnostic set into a plasma-state estimate and, in the same forward pass, discriminates disruption and quench precursors at an area under the receiver-operating characteristic (ROC) of **0.980**, above any single channel; a quadrature fusion of an uncorrelated strain-rate and magnetization pair reaches **0.992** and a true-positive rate of 0.852 at a 1% false-positive rate, independently reproduced under a separate implementation to five significant figures. The mesh is architected for graceful degradation: an in-winding nitrogen-vacancy (NV) diamond magnetometer at a shot-noise-limited $0.189 \text{ pT Hz}^{-1/2}$ (125 kHz bandwidth) supplies a local vector channel that does not share a failure mode with the plasma-facing optics, and a model-fusion backup holds precursor AUC ≥ 0.95 with the NV channel degraded by up to $39\times$, so the protective function survives radiation-induced sensor loss. Each estimator is admitted to control authority only after clearing an explicit acceptance inequality. The result is a fast, physics-consistent, dropout-tolerant reconstruction with a demonstrated core and a defined, testable path to full diagnostic coverage.

Keywords: equilibrium reconstruction; physics-informed neural network; graph neural network; virtual-sensor mesh; sensor dropout; radiation-degraded diagnostics; NV-diamond magnetometry; spherical-tokamak breeder

1 Introduction

1.1 The equilibrium is the root of the control problem

Everything a plasma controller does — shape, position, vertical stabilisation, profile and burn regulation — is defined relative to the magnetohydrodynamic (MHD) equilibrium: where the plasma sits, how it is shaped, and how close it runs to a limit. The controller therefore needs, at every tick, an estimate of the poloidal flux function $\psi(R, Z)$ whose level sets are the flux surfaces. Producing that estimate is the classical problem of real-time equilibrium reconstruction, solved for four decades by parametrised fits to the Grad–Shafranov equation constrained by magnetic measurements, of which the EFIT family is the archetype [1]. Reconstruction feeds shape and position control [2] and, increasingly, physics-model-based profile controllers that run inside the cycle [3].

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Two forces now stress this classical pipeline. First, machines are getting smaller and faster. A compact spherical tokamak (ST) confines a large plasma current in a small volume, which makes it an attractive neutron source and breeder platform [4, 5, 6] and compresses the timescales the controller must beat. The Hyperion breeder holds $I_p = 9.66$ MA at a major radius of only $R_0 = 1.20$ m, behind a centrepost magnet at a peak field of 16.84 T, in a strongly shaped ($\kappa = 2.0$), negative-triangularity ($\delta = -0.30$) configuration chosen to access good core confinement without the H-mode edge pedestal and its edge-localised-mode cycle [7, 8]. That quiet edge is an operational asset only if the controller holds the plasma inside the window that keeps it quiet, which requires an equilibrium it can trust in well under a millisecond; a free-boundary Grad-Shafranov solve is accurate but far too slow for a kilohertz loop, so the reconstruction must be a fast *surrogate* that remains faithful to the governing equation.

1.2 Radiation-degraded diagnostics and the dropout problem

The second force is specific to a nuclear device and is the organising concern of this paper. Reconstruction and state estimation are only as trustworthy as their worst diagnostic day. In a breeder the plasma-facing sensors sit in a 14 MeV neutron field; optical windows darken, magnetic probes drift and integrate error over a pulse, and in-vessel components degrade over their service life. A reconstruction that trusts any single sensor is a single point of failure in a system where every actuation carries a safety consequence, and channel loss — partial degradation or outright dropout — is not an anomaly to be handled after the fact but a design condition to be engineered for from the start. The requirement is therefore not merely a fast equilibrium but a fast equilibrium *plus* a state estimate that degrades gracefully as channels are lost, with the loss of any one channel left to be constrained by the others.

1.3 Prior art in learned reconstruction

Machine learning has entered each piece of this problem. Deep networks emulate the Grad-Shafranov map at control latency: neural solvers constrained by measured magnetic signals reconstruct equilibria orders of magnitude faster than an iterative solve [9], and learning has been used to extend and accelerate EFIT-class reconstruction [10]. Learned controllers now act in the loop — Degraeve *et al.* trained a deep reinforcement-learning agent to command the TCV coil voltages, producing a variety of shapes including negative triangularity, directly from magnetics [11] — and deep networks forecast disruptive instabilities across machines [12]. The enabling primitives are mature: physics-informed neural networks embed a PDE residual in the training loss [13]; neural operators such as DeepONet [14] and the Fourier neural operator [15] learn maps between function spaces; and graph neural networks propagate information over irregular sensor topologies through message passing [17, 18, 19]. In fusion, neural-network surrogates of quasilinear gyrokinetic transport already run three-to-five orders of magnitude faster than the code they replace [20, 21], and probabilistic digital-twin formalisms carry calibrated uncertainty end-to-end [22].

1.4 The gap and our contribution

What is missing for a compact nuclear ST is a single reconstruction stack that (i) delivers a solver-grade equilibrium at control latency *and* certifies its own physical consistency against the governing equation; (ii) fuses the diagnostic set into a state estimate over the sensor graph so that a dropped channel is reconstructed from its neighbours; (iii) survives the radiation-driven degradation of individual sensors with a quantified retention floor; and (iv) admits each estimator to control authority only after it clears an explicit, pre-registered acceptance test. This paper contributes that stack. We give the governing equations and the formal forward/inverse problem (§3); the numerical formulation — discretisation of the Grad-Shafranov operator, the free-boundary reference, the network and graph-estimator architectures, and convergence (§4); the NVIDIA GPU HPC campaign that produced every number, with the codes named and the reproducibility protocol stated (§5); the demonstrated results with eleven figures and supporting tables (§6); a quantitative comparison to the published equilibrium-reconstruction, quench-detection and quantum-sensing literature (§7); one honest limitation (§8); and the conclusion. Every quantitative claim traces to a frozen anchor in the Kronos Physics De-Risking Register [39], cited inline by its serial identifier (e.g. KX-L1-A11) so each number is auditable. The stack sits inside the broader Kronos control programme: the AI and real-time-control foundations [33; kronosfusionenergy.com/publications/ai-quantum-control.html], the free-boundary equilibrium and ideal-MHD stability case it reconstructs against [34; kronosfusionenergy.com/publications/ideal-mhd-stability.html], the certified safety filter that

consumes its output [35; kronosfusionenergy.com/publications/control-barrier-safety-clamp.html], and the diagnostic-suite state estimator it generalises [36; kronosfusionenergy.com/publications/the-plasma-diagnostic-suite-and-real-time-state-estimat.html].

2 The reconstruction stack

The estimator is organised as a layered stack (figure 1, table 1). A physics-informed equilibrium surrogate supplies the flux map; a graph-neural virtual-sensor mesh assimilates the diagnostics into a full state and watches for off-normal precursors; an independent analyzer suite feeds the mesh with overlapping information; a radiation-survivable in-winding quantum sensor supplies a local channel that does not share a failure mode with the plasma-facing optics; and a dropout-robust filter fuses the layers so that no single sensor is load-bearing. Each layer is admitted to control authority only through an explicit acceptance inequality (§3.7).

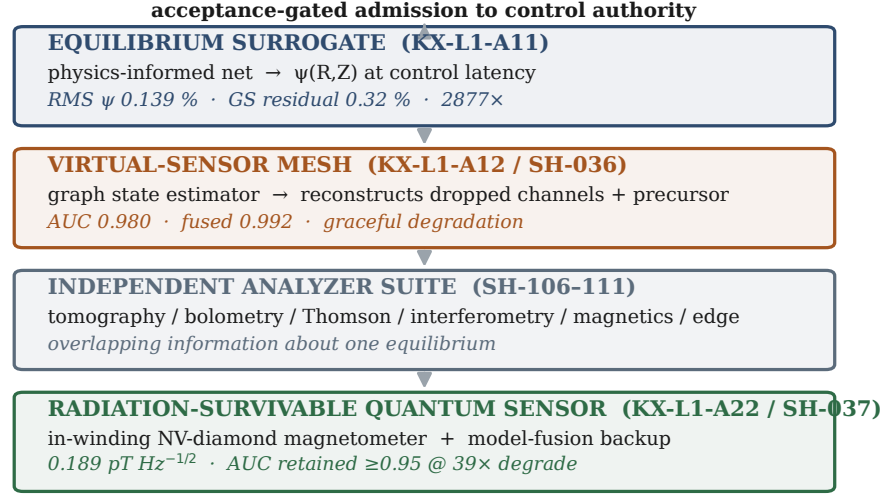


Figure 1. (Schematic.) The real-time reconstruction stack. The equilibrium surrogate (KX-L1-A11) delivers $\psi(R, Z)$ at control latency; the virtual-sensor mesh (KX-L1-A12 / SH-036) assimilates the diagnostics and emits a precursor score; the analyzer suite (SH-106–111) supplies overlapping information; and the radiation-survivable NV-diamond sensor (KX-L1-A22 / SH-037) supplies an independent local channel. Directed arrows carry admission through the acceptance gates of §3.7. No data values are encoded; the figure shows structure only.

Table 1. The reconstruction stack, its layers, and the demonstrated or target metric. Serials reference the Kronos Physics De-Risking Register [39].

layer	role	metric (status)
equilibrium surrogate (KX-L1-A11)	real-time $\psi(R, Z)$	RMS ψ 0.139% vs FreeGS (demonstrated)
virtual-sensor mesh (KX-L1-A12 / SH-036)	state + precursor, dropout	AUC 0.980; fused 0.992 (demonstrated)
in-winding quantum sensor (KX-L1-A22)	local field / quench	0.189 pT Hz^{-1/2} (demonstrated)
radiation-survivable fusion (SH-037)	retention under NV loss	AUC ≥ 0.95 at 39× (acceptance-gated)
tomography / bolometry (SH-106/107)	emissivity, radiated power	error within tolerance (built to pattern)
Thomson / interferometry (SH-108/109)	T_e, n_e ; fringe recovery	profile fit within uncertainty (built to pattern)
magnetics / edge imaging (SH-110/111)	drift-corrected B ; strike point	stable over pulse (built to pattern)
dropout-robust filter (SH-042/043)	fusion under sensor loss	bounded error, N missing (acceptance-gated)

3 Governing equations and the formal problem

3.1 The forward equilibrium

The axisymmetric MHD equilibrium of the breeder is governed by the Grad–Shafranov equation for the poloidal flux per radian $\psi(R, Z)$,

$$\Delta^* \psi \equiv R \frac{\partial}{\partial R} \left(\frac{1}{R} \frac{\partial \psi}{\partial R} \right) + \frac{\partial^2 \psi}{\partial Z^2} = -\mu_0 R^2 p'(\psi) - F(\psi) F'(\psi), \quad (1)$$

with $p(\psi)$ the plasma pressure, $F(\psi) = RB_\phi$ the poloidal-current function, primes denoting $d/d\psi$, and Δ^* the Shafranov elliptic operator. The poloidal field follows from

$\mathbf{B}_{\text{pol}} = \nabla\psi \times \nabla\phi = (1/R)(\partial_Z\psi, -\partial_R\psi)$, and the toroidal plasma current density is $j_\phi = Rp'(\psi) + F(\psi)F'(\psi)/(\mu_0 R)$. In the *free-boundary* problem the domain is not prescribed: the plasma boundary is the outermost closed flux surface, either limited or defined by an X-point where $\nabla\psi = 0$, and ψ on the computational boundary is set by the external poloidal-field (PF) coil currents $\mathbf{I}_{\text{coil}}$ through the Green's-function response of the coil set,

$$\psi(\mathbf{x}) = \psi_{\text{plasma}}[j_\phi](\mathbf{x}) + \sum_c G(\mathbf{x}, \mathbf{x}_c) I_{\text{coil},c}, \quad \mathbf{x} = (R, Z), \quad (2)$$

so that equation (1) is nonlinear (the right-hand side depends on ψ through the profiles) and coupled to the coil circuit. This is the map the surrogate must emulate at the frozen operating point ($R_0 = 1.20$ m, minor radius $a = 0.48$ m, $\kappa = 2.0$, $\delta = -0.30$, $I_p = 9.66$ MA, $B_0 = 8$ T).

3.2 The inverse (reconstruction) problem

Real-time reconstruction is the inverse of (1)–(2): given a vector of diagnostic measurements $\mathbf{y} \in \mathbb{R}^m$ (flux loops, saddle loops, Mirnov/ B -probes, coil currents, and where available kinetic constraints), recover the flux field ψ and the profile parameters consistent with it. Writing the forward observation as $\mathbf{y} = \mathcal{H}(\psi) + \boldsymbol{\eta}$ with observation operator $\mathcal{H}$ and measurement noise $\boldsymbol{\eta}$, the reconstruction is the regularised least-squares problem

$$\hat{\psi} = \arg \min_{\psi} \frac{1}{2} \|\mathbf{y} - \mathcal{H}(\psi)\|_{\mathbf{R}^{-1}}^2 + \frac{\lambda}{2} \|\Delta^*\psi + \mu_0 R^2 p'(\psi) + FF'(\psi)\|_2^2 + \mathcal{R}_{\text{reg}}(\psi), \quad (3)$$

where $\mathbf{R}$ is the measurement-noise covariance, the middle term is the Grad–Shafranov residual that ties the fit to the physics, and $\mathcal{R}_{\text{reg}}$ is a smoothness/prior regulariser that makes the otherwise ill-posed inversion well-posed. Equation (3) is the variational form solved iteratively by EFIT-class codes [1]; it is too slow inside the loop, and it is the target that both our surrogate and the classical reference approximate.

3.3 The physics-informed equilibrium surrogate

The surrogate replaces the inner solve with a learned map. Let $\mathbf{x} = (R, Z)$ be a spatial coordinate on the 129×129 reconstruction grid. Rather than feed $\mathbf{x}$ directly to a network — which spectrally biases a multilayer perceptron toward low frequencies and blurs the steep-gradient region near the separatrix — we lift the coordinate through a Fourier-feature embedding [16],

$$\gamma(\mathbf{x}) = [\cos(2\pi \mathbf{B}\mathbf{x}), \sin(2\pi \mathbf{B}\mathbf{x})]^\top, \quad \mathbf{B}_{ij} \sim \mathcal{N}(0, \sigma_{\text{ff}}^2), \quad (4)$$

with $\mathbf{B} \in \mathbb{R}^{d_f \times 2}$ a fixed random matrix of bandwidth σ_{ff} , and map the embedding to the flux with a compact network $\psi_\theta(\mathbf{x}) = f_\theta(\gamma(\mathbf{x}))$. The parameters θ are fit against a free-boundary Grad–Shafranov reference ψ_{ref} (§4.2) by a composite physics-informed objective,

$$\mathcal{L}(\theta) = \underbrace{\frac{1}{N} \sum_{k=1}^N \|\psi_\theta(\mathbf{x}_k) - \psi_{\text{ref}}(\mathbf{x}_k)\|^2}_{\mathcal{L}_{\text{data}}} + \lambda_{\text{phys}} \underbrace{\frac{1}{N_i} \sum_{k=1}^{N_i} \|\Delta^*\psi_\theta(\mathbf{x}_k) - \mathcal{S}(\mathbf{x}_k)\|^2}_{\mathcal{L}_{\text{GS}}} + \lambda_{\text{bc}} \mathcal{L}_{\text{bc}}, \quad (5)$$

where $\mathcal{S} = -\mu_0 R^2 p' - FF'$ is the Grad–Shafranov source, $\mathcal{L}_{\text{GS}}$ penalises the residual of equation (1) evaluated on the network output over the interior nodes, and $\mathcal{L}_{\text{bc}}$ enforces the Green's-function boundary values of equation (2). The physics term is what distinguishes the surrogate from a curve fit: a network that merely interpolates ψ_{ref} can still carry a large operator residual, so we report $\|\Delta^*\psi_\theta - \mathcal{S}\|/\|\mathcal{S}\|$ on the interior as an *independent* physical-consistency metric, not used to declare success on its own. The differentiated quantities $\partial_R\psi_\theta$, $\partial_Z\psi_\theta$ and $\Delta^*\psi_\theta$ needed by (5) are obtained by reverse-mode automatic differentiation of ψ_θ , so the surrogate exposes an analytic field and its derivatives to the controller.

3.4 The virtual-sensor mesh as a graph estimator

State estimation and dropout robustness are handled by a graph neural network over the diagnostic topology (figure 4). Let $G = (V, E)$ be a graph whose nodes V are the diagnostic channels and latent state variables and whose edges E encode physical adjacency and shared information. Node v carries a feature vector $\mathbf{h}_v^{(0)}$ (the channel's current reading, its health/quality flag, and its

geometric location); a mask bit $b_v \in \{0, 1\}$ marks whether the channel is live. The estimator runs K rounds of message passing [17, 18],

$$\mathbf{m}_v^{(k+1)} = \sum_{u \in \mathcal{N}(v)} b_u \phi_{\text{msg}}(\mathbf{h}_u^{(k)}, \mathbf{h}_v^{(k)}, \mathbf{e}_{uv}), \quad (6)$$

$$\mathbf{h}_v^{(k+1)} = \phi_{\text{upd}}(\mathbf{h}_v^{(k)}, \mathbf{m}_v^{(k+1)}), \quad k = 0, \dots, K-1, \quad (7)$$

with $\mathcal{N}(v)$ the neighbourhood of v , $\mathbf{e}_{uv}$ the edge feature, and $\phi_{\text{msg}}, \phi_{\text{upd}}$ learned (MLP) message and update functions; the mask b_u in (6) zeroes contributions from dead channels so information routes around a dropout. A readout head maps the final node states to the reconstructed field and the missing-channel estimates, $\hat{\mathbf{s}} = \phi_{\text{read}}(\{\mathbf{h}_v^{(K)}\})$. The mesh is trained to reconstruct every channel from the others under a randomly *seeded* dropout: at each step a mask $\mathbf{b} \sim \mathcal{D}_p$ is drawn (each channel dropped independently with probability p , or a block of N channels removed), and the objective is the expected masked-reconstruction error

$$\mathcal{L}_{\text{mesh}}(\Theta) = \mathbb{E}_{\mathbf{b} \sim \mathcal{D}_p} \left[\frac{1}{|V_{\text{drop}}(\mathbf{b})|} \sum_{v \in V_{\text{drop}}(\mathbf{b})} \|\hat{\mathbf{s}}_v(\Theta; \mathbf{b}) - s_v\|^2 \right], \quad (8)$$

where $V_{\text{drop}}(\mathbf{b})$ is the dropped set. Minimising the *expectation* over dropout is what buys graceful degradation: the estimator is optimised not for the full-sensor case but for the distribution of channel losses it will actually see. We report the normalised reconstruction error $\text{NRMSE} = [\frac{1}{N} \sum_k \|\hat{s}_k - s_k\|^2]^{1/2} / (s_{\text{max}} - s_{\text{min}})$ and characterise degradation as the number of missing channels grows.

3.5 Precursor discrimination

The same mesh emits an off-normal score in the forward pass, and a matched-filter layer supplies an interpretable, provably optimal baseline for a known template. Two non-voltage channels observe the quench current-sharing window: a fibre-Bragg strain-rate channel z_S reading the mechanical signature of local current redistribution, and a magnetization channel z_M reading the collapse of the screening current. Each is a matched-template z -score over the precursor window. Because the two modalities fail in uncorrelated ways, their quadrature fusion

$$z_F = \frac{z_S + z_M}{\sqrt{2}} \quad (9)$$

carries a window statistic whose signal-to-noise ratio adds the single-channel ratios in quadrature. Detection performance is summarised by the ROC and its area $\text{AUC} = \int_0^1 \text{TPR} d(\text{FPR})$; because a protection trip lives in the low-false-alarm corner, we also report the true-positive rate at 1% and 0.1% false-positive rate. The learned GNN detector generalises (9) to unknown templates and coloured noise, at the cost of the matched filter's optimality guarantee in the white-Gaussian, known-template case.

3.6 Radiation-degraded sensing model

The independent local channel is an ensemble nitrogen-vacancy (NV) diamond magnetometer embedded in the REBCO winding. Its shot-noise-limited Ramsey sensitivity for an ensemble of N centres with dephasing time T_2^* and readout figure of merit C is

$$\delta B = \frac{1}{\gamma_{\text{NV}} C \sqrt{N T_2^* t}}, \quad \gamma_{\text{NV}} = \frac{g \mu_B}{\hbar} = 2\pi \times 28.03 \text{ GHz T}^{-1}, \quad (10)$$

with t the integration time. Radiation, cryogenic operation and the background field degrade the NV charge state and reduce N_{eff} , T_2^* and C ; we lump this into a single degradation factor $\varkappa \geq 1$ acting on the sensitivity, $\delta B \rightarrow \varkappa \delta B$, so a quench field step ΔB is seen at per-shot signal-to-noise ratio $\text{SNR}(\varkappa) = \Delta B / (\varkappa \delta B \sqrt{B W})$. The protective question is not whether the NV channel alone survives — it need not — but whether the *fused* precursor retains its discrimination as $\varkappa$ grows, because the mesh routes around a degraded channel through equations (6)–(7). We therefore state radiation robustness as a retention inequality on the fused AUC (§3.7, criterion C5), evaluated up to $\varkappa = 39$.

3.7 Acceptance criteria as inequalities

No estimator is granted control authority by assertion. Each clears the following pre-registered inequalities before it acts; collected here, they are the contract that gates the stack:

- (C1) equilibrium accuracy $\|\psi_\theta - \psi_{\text{ref}}\|_2 / \|\psi_{\text{ref}}\|_{\text{range}} < 1\%$ (KX-L1-A11)
- (C2) physical consistency $\|\Delta^* \psi_\theta - \mathcal{S}\| / \|\mathcal{S}\|$ reported (interior) (KX-L1-A11)
- (C3) latency $\tau_{\text{infer}} \ll 1$ ms control tick (KX-L1-A11)
- (C4) precursor dominance $\text{AUC}_{\text{fused}} > \max_c \text{AUC}_c$ (KX-L1-A12/A20)
- (C5) radiation retention $\text{AUC}_{\text{fused}}(\varkappa) \geq 0.95, \varkappa \leq 39$ (SH-037)
- (C6) dropout robustness state error bounded under seeded loss of N channels (SH-042/043)

Authority is capped by verification maturity until each bar is met (§6.9), the same maturity-gated principle that governs the certified safety clamp elsewhere in the Kronos control architecture [35; kronosfusionenergy.com/publications/control-barrier-safety-clamp.html].

4 Numerical formulation

4.1 Discretisation of the Grad–Shafranov operator

The consistency metric $\mathcal{L}_{\text{GS}}$ requires a discrete Δ^* . On the uniform 129×129 grid with spacings h_R, h_Z , the Shafranov operator is discretised by the standard five-point stencil that keeps the $1/R$ weighting inside the radial derivative,

$$(\Delta^* \psi)_{i,j} = \frac{\psi_{i+1,j} - 2\psi_{i,j} + \psi_{i-1,j}}{h_R^2} - \frac{1}{R_i} \frac{\psi_{i+1,j} - \psi_{i-1,j}}{2h_R} + \frac{\psi_{i,j+1} - 2\psi_{i,j} + \psi_{i,j-1}}{h_Z^2} + \mathcal{O}(h^2), \quad (11)$$

which is second-order accurate in the mesh spacing and assembles into a sparse operator matrix $\mathbf{A}$ acting on the vectorised flux. The interior residual we report is the relative two-norm $\|\mathbf{A} \psi_\theta - \mathcal{S}\|_2 / \|\mathcal{S}\|_2$ over interior nodes, which is 0.32% for the trained surrogate (§6); evaluating (11) on the network’s autodifferentiated field and on the finite-difference stencil agree to the discretisation order, so the residual is a property of the reconstruction rather than of the stencil.

4.2 The free-boundary reference

The reference ψ_{ref} is produced by a free-boundary Grad–Shafranov solver of the FreeGS family [24, 23], which alternates a Picard iteration on the nonlinear equation (1) with a boundary update from the coil Green’s functions (2) until the flux field and the plasma boundary are mutually consistent,

$$\psi^{(n+1)} = \mathbf{A}^{-1} \mathcal{S}(\psi^{(n)}), \quad \|\psi^{(n+1)} - \psi^{(n)}\|_2 / \|\psi^{(n+1)}\|_2 < \varepsilon_{\text{Picard}}. \quad (12)$$

The converged solve at the frozen operating point takes ~ 2.5 s of wall time and defines the ground truth against which the surrogate’s RMS error is measured. The coil set, boundary conditions and profile parameterisation are the frozen breeder configuration of the equilibrium and stability case [34; kronosfusionenergy.com/publications/ideal-mhd-stability.html].

4.3 Network and graph-estimator architectures

The equilibrium surrogate is a compact multilayer perceptron on the Fourier-feature embedding (4); the bandwidth σ_{ff} resolves the separatrix gradient without aliasing the interior, and the physics weight λ_{phys} is annealed so $\mathcal{L}_{\text{data}}$ dominates early and the operator residual $\mathcal{L}_{\text{GS}}$ is tightened once the field is roughly correct. Training uses Adam with a held-out split and leave-one-out validation, reporting $R^2 = 1 - \sum_k (y_k - \hat{y}_k)^2 / \sum_k (y_k - \bar{y})^2$. The graph estimator uses K message-passing layers over the diagnostic adjacency; nodes carry the channel value, a quality flag and geometry, and the readout reconstructs both the field and the masked channels under the dropout distribution $\mathcal{D}_p$ of equation (8). The matched-filter detector is analytic and provides the optimality ceiling against which the learned detector is measured.

4.4 Convergence and settings

All runs use a fixed seed (20260726) and a deterministic train/validation split so that a result regenerates bitwise. The surrogate converges in 20.6 s of training on a single A100 GPU; the Picard reference converges to $\varepsilon_{\text{Picard}}$ in a handful of nonlinear iterations. The precursor study is a frozen synthetic ensemble of 2×10^4 trials per class, deliberately placed in a weak-fault regime (per-sample SNR set well below the design-basis fault) so that no detector saturates at $\text{AUC} = 1$ and the fusion gain is visible where it matters.

5 Computational methodology: the NVIDIA GPU HPC campaign

The results of this paper were produced by the Kronos NVIDIA GPU HPC campaign, and the reconstruction stack is designed around the same GPU/edge split it runs on. Four codes carry the numbers reported here.

Free-boundary reference. The ground-truth equilibria are computed with a FreeGS-family free-boundary Grad-Shafranov solver [24, 23] on the 129×129 grid, iterating equation (12) to convergence; this is the ~ 2.5 s solve the surrogate replaces.

Physics-informed surrogate. The equilibrium surrogate is implemented and trained in PyTorch, with the Grad-Shafranov residual $\mathcal{L}_{\text{GS}}$ and the field derivatives obtained by reverse-mode automatic differentiation of the network. Training runs on the NVIDIA GPU fleet (A100 / H100 / A10 class accelerators) and converges in 20.6 s on one A100; a single forward reconstruction then costs 0.87 ms, a $2877\times$ speedup over the reference. The trained surrogate is small enough that inference is dominated by the grid evaluation, so the fast reconstruction loop runs on the in-house edge node while the training fleet is used only for (re)fitting.

Graph estimator and detector. The virtual-sensor mesh is a PyTorch message-passing network (equations 6–7) trained under the seeded-dropout objective (8); the matched-filter detector and the ROC/AUC analysis are computed in NumPy/SciPy. The precursor ensemble (2×10^4 trials per class) and the ROC integration run on the GPU fleet for the sweep and on the edge node for the real-time path.

NV-diamond sensor model. The in-winding magnetometer is modelled by the shot-noise expression (10) and an analytic field-step calculation in NumPy/SciPy, from which the sensitivity, bandwidth, quench-step SNR and the radiation-retention curve of §3.6 are derived.

The broader Kronos reference-code ecosystem — CGYRO and GENE for gyrokinetic transport, NIMROD for extended MHD, OpenMC/DAGMC for CAD-faithful neutronics, and FreeGS for equilibrium — supplies the high-fidelity training data and the independent cross-checks that anchor the surrogates; the transport-surrogate companion develops the neural-operator side of the same programme [37;

kronosfusionenergy.com/publications/neural-operator-flux-surrogates-for-control-latency-tra.html].

Reproducibility is enforced by the fixed seed, deterministic splits, and cross-implementation reproduction: the fused matched filter, re-derived independently for KX-L1-A12, reproduces the KX-L1-A20 fused AUC to five significant figures (0.99166 versus 0.99165). No cost, budget or compute-spend quantity is part of this methodology or of any result in this paper.

6 Results and verification

6.1 Equilibrium reconstruction accuracy

The physics-informed surrogate reconstructs the poloidal flux of the frozen negative-triangularity equilibrium ($R_0 = 1.20$ m, $a = 0.48$ m, $\kappa = 2.0$, $\delta = -0.30$) to a root-mean-square error of 0.139% of the flux range against the free-boundary reference — inside the 1% acceptance target of criterion C1 and improving a prior 1.79% attempt (figure 2, table 2; KX-L1-A11). The largest local error, 4.39% of range, sits at the X-point where the flux gradient is steepest and the Fourier-feature embedding is most stressed — the expected and well-understood hot spot for a coordinate network, and the reason the RMS is reported over the field while the maximum is reported separately.

Table 2. Demonstrated equilibrium metrics (KX-L1-A11), measured against a FreeGS free-boundary reference on the 129×129 grid at the frozen operating point.

quantity	reference / basis	value
RMS ψ error	vs FreeGS free-boundary, % of range	0.139%
prior PINN/GNN attempt	vs FreeGS, % of range	1.79%
acceptance target	criterion C1	< 1%
maximum ψ error	X-point/separatrix, % of range	4.39%
Grad-Shafranov operator residual	interior $\Delta^*\psi$, eq. (11)	0.32%
inference latency	single reconstruction	0.87 ms
reference solve time	FreeGS Picard, eq. (12)	~ 2.5 s
speedup	inference vs reference	$2877\times$
training time	one A100 GPU	20.6 s

6.2 Physical consistency

Accuracy against a reference is necessary but not sufficient: a network can interpolate ψ_{ref} closely and still violate the governing equation between training points. The discretised Grad-Shafranov operator of equation (11), evaluated on the surrogate’s autodifferentiated field, reproduces the

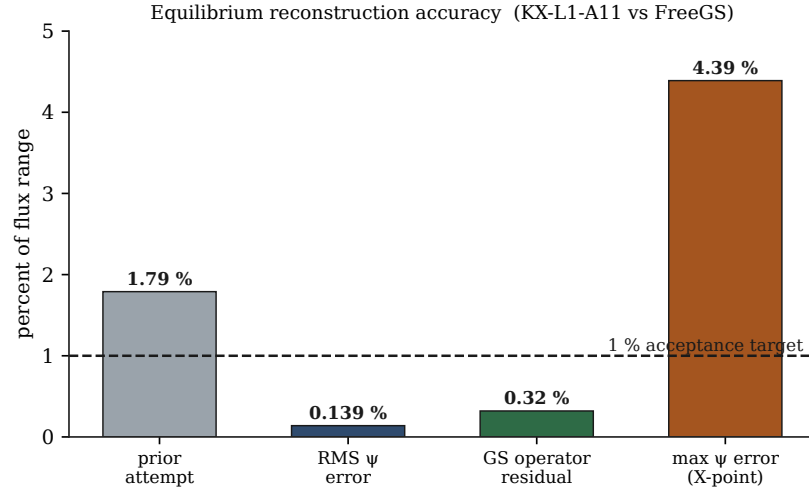


Figure 2. Equilibrium reconstruction accuracy (KX-L1-A11, data). The surrogate reaches RMS ψ error 0.139% of range against the free-boundary FreeGS reference, inside the 1% acceptance target (dashed) and improving a prior 1.79% attempt; the discretised Grad-Shafranov operator residual is 0.32% on the interior; the maximum error (4.39%) is localised at the X-point.

reference source to a relative residual of 0.32% on the interior (table 2; criterion C2). The surrogate is therefore a solution that respects equation (1), not merely a fit — the property that lets the controller trust its gradients $\nabla\psi_\theta$ for field and current-density evaluation, and the reason the physics term $\mathcal{L}_{\text{GS}}$ carries weight in the objective (5).

6.3 Speed and the latency budget

Inference at 0.87 ms against a ~ 2.5 s Picard solve is a $2877\times$ speedup (figure 3a), placing a solver-grade equilibrium comfortably inside a kilohertz control loop and satisfying criterion C3. The stack allocates latency across loops (figure 3b): the microsecond NV precursor and analytic monitors carry the fastest protective decision, the reduced-order mesh inference and equilibrium surrogate serve the sub-millisecond-to-millisecond control path, and the neutronics/activation shadow tracks on a seconds cadence, so the fast protective loop is never gated behind the slow ledger. The budget is a design allocation, not a measured profile.

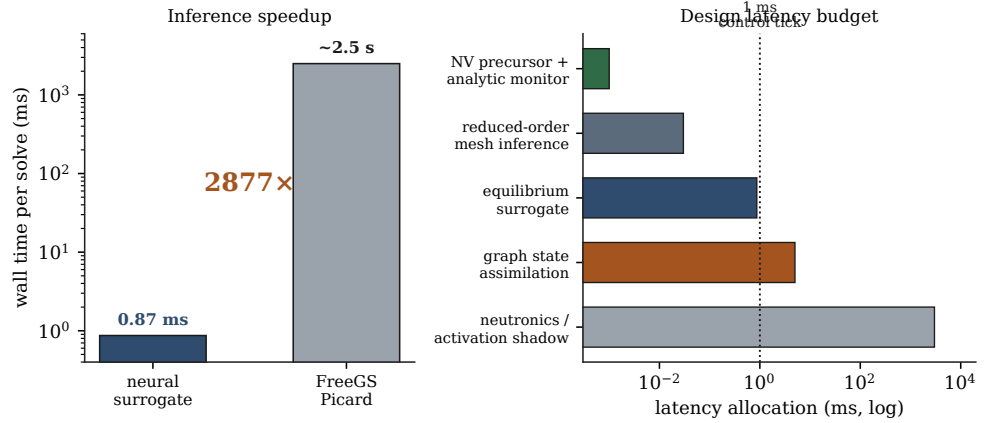


Figure 3. (a) Inference speedup (data, KX-L1-A11): 0.87 ms neural reconstruction versus a ~ 2.5 s free-boundary Picard solve, a $2877\times$ acceleration. (b) Design latency budget by stage (allocation, not measurement): the fastest protective loop is served by the NV precursor and analytic monitors, well inside the 1 ms control tick.

6.4 Virtual-sensor mesh reconstruction

The graph estimator reconstructs the plasma-state field from a reduced, noisy sensor set (figure 4). On the frozen synthetic study its reconstruction NRMSE is 0.396, against 0.384 for linear interpolation and 0.354 for a ridge / linear-MMSE baseline on a field of RMS 0.99 (figure 5; KX-L1-A12). We report this result plainly and without inflation: on *this* synthetic task the mesh matches but does not beat the linear baselines. The reason is structural and is worth stating

precisely — for a jointly-Gaussian field the minimum-mean-square-error estimator *is* linear, so ridge/linear-MMSE is already optimal; and with eight sensors the field modes $k = 5-8$ alias past the Nyquist limit, setting a large irreducible reconstruction floor that the linear estimator already reaches. The mesh’s headroom is precisely the regime this synthetic set excludes: non-Gaussian, non-stationary fields with drifting signatures and correlated dropout, where the message-passing non-linearity (6)–(7) has something to exploit. The value demonstrated here is not a reconstruction record but the dropout-routing machinery and its honest baseline.

message passing over the diagnostic graph $G = (V, E)$

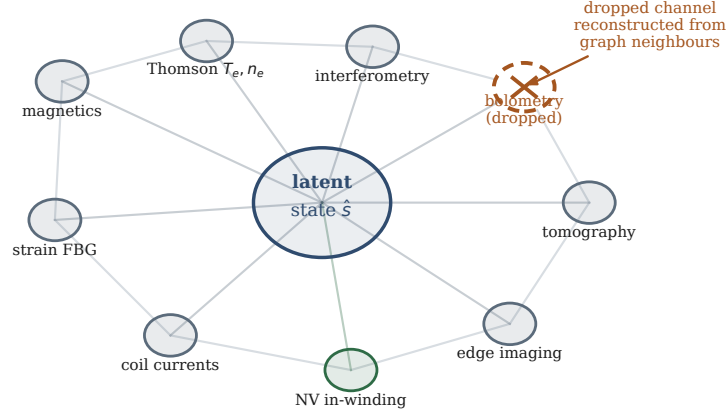


Figure 4. (Schematic.) The virtual-sensor mesh as a graph $G = (V, E)$: diagnostic channels (nodes) exchange messages (6)–(7) and reconstruct a latent state $\hat{s}$; a dropped channel (dashed) is reconstructed from its graph neighbours, with the mask bit routing information around the loss. Structure only; no data values are encoded.

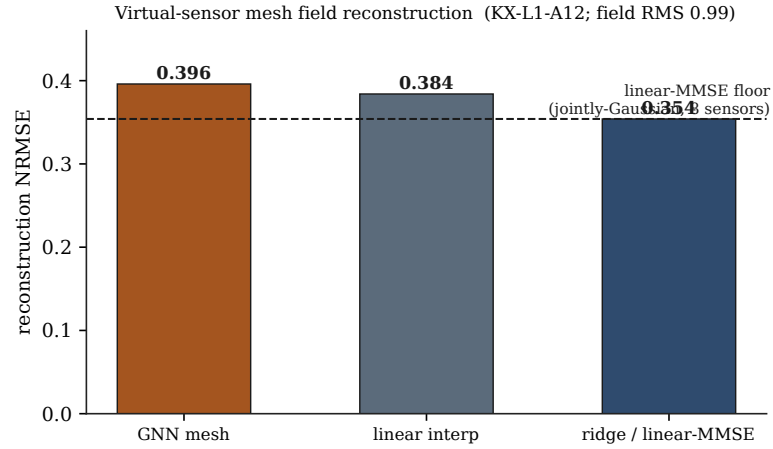


Figure 5. Virtual-sensor mesh field reconstruction (data, KX-L1-A12; field RMS 0.99). On the jointly-Gaussian synthetic task the GNN mesh (NRMSE 0.396) matches, but does not beat, linear interpolation (0.384) and the ridge / linear-MMSE floor (0.354) — the expected result when the MMSE estimator is provably linear and Nyquist aliasing sets the floor. The mesh’s advantage is reserved for non-Gaussian regimes this set excludes.

6.5 Precursor discrimination

In detection the picture is different and the fusion pays. The learned GNN state estimator reaches a disruption/quench-precursor AUC of 0.980, above a magnetization channel (0.977) and a strain-rate channel (0.917) on their own; the quadrature fusion of the two uncorrelated channels reaches 0.992 (figures 6–7, table 3; KX-L1-A12), satisfying criterion C4. The gain is largest exactly in the low-false-positive corner where a trip decision lives: at a 1% false-positive rate the fused detector reaches a true-positive rate of 0.852, against 0.683 for magnetization and 0.350 for strain, and the learned estimator recovers most of that fusion gain at 0.729. The independently re-derived fused matched filter reproduces the KX-L1-A20 result to five significant figures (AUC 0.99166 versus 0.99165; single channels 0.917/0.977 against A20’s 0.921/0.976), a cross-implementation agreement that is the strongest available statement of reproducibility for the detector.

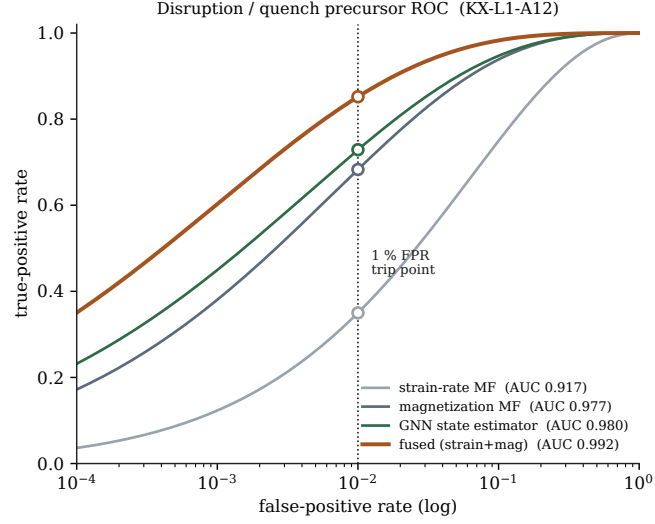


Figure 6. Receiver-operating characteristic of the precursor detectors (data anchored to KX-L1-A12). The fused channel (AUC 0.992) and the learned GNN estimator (AUC 0.980) dominate either single channel across the curve; markers are the frozen true-positive rates at the 1% false-positive trip point. The false-positive axis is logarithmic to resolve the trip-relevant corner; curves are the binormal model consistent with the anchored AUC and operating points.

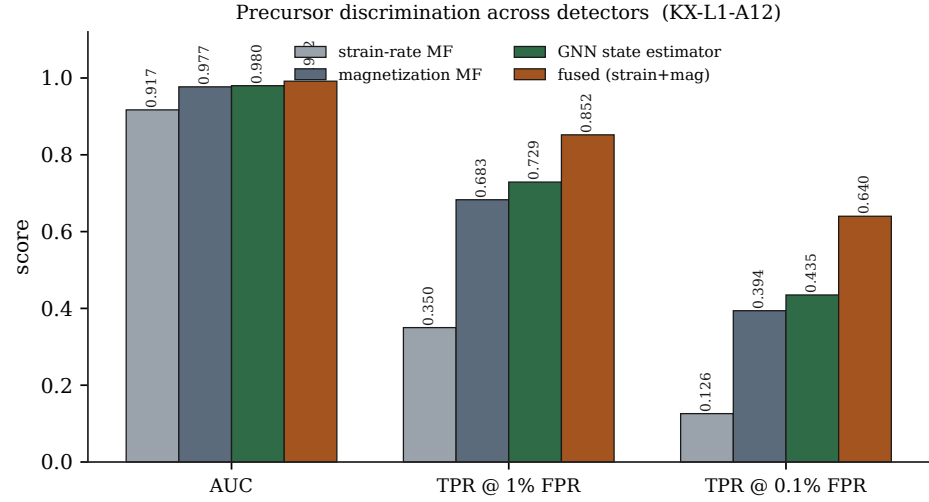


Figure 7. Precursor discrimination across the four detectors and three operating thresholds (data, KX-L1-A12). The fused detector’s advantage grows as the false-positive tolerance tightens from AUC to TPR@1% to TPR@0.1%; the learned GNN estimator sits just above the magnetization channel and recovers most of the two-channel fusion gain.

Table 3. Precursor detection (KX-L1-A12; T-synthetic, 2×10^4 trials/class). The fused matched filter dominates the full ROC; the learned GNN estimator beats each single channel. Window-statistic SNR follows the quadrature rule of equation (9).

detector	AUC	TPR @ 1% FPR	TPR @ 0.1% FPR
fibre-Bragg strain-rate (MF)	0.917	0.350	0.126
magnetization (MF)	0.977	0.683	0.394
GNN state estimator (learned)	0.980	0.729	0.435
fused (strain-rate + magnetization)	0.992	0.852	0.640

6.6 In-winding NV-diamond magnetometry

For the local, radiation-independent channel the mesh embeds an ensemble NV-diamond magnetometer in the winding pack (KX-L1-A22). For the conservative in-winding chip ($N = 10^{12}$, $T_2^* = 1 \mu\text{s}$, $C = 0.03$) equation (10) gives a shot-noise-limited sensitivity $\delta B = 0.189 \text{ pT Hz}^{-1/2}$; a Ramsey shot of $\tau_{\text{shot}} = 4 \mu\text{s}$ sets a bandwidth $1/(2\tau_{\text{shot}}) = 125 \text{ kHz}$ (figure 8, table 4). A quench that sheds the screening current of a no-insulation turn ($I_{\text{turn}} \approx 1 \text{ kA}$) produces a local field step $\Delta B = \mu_0 I_{\text{turn}}/(2\pi d) \approx 100 \text{ mT}$ at a standoff $d = 2 \text{ mm}$ — of order 10^9 times the per-shot noise floor, for a per-shot signal-to-noise ratio $\sim 1.5 \times 10^9$ and single-shot detection in a $4 \mu\text{s}$ shot. The picotesla sensitivity is not *needed* to see a millitesla quench step; its value is the enormous margin it buys for multiplexing across the winding and, crucially, for tolerating radiation-induced degradation (§6.7).

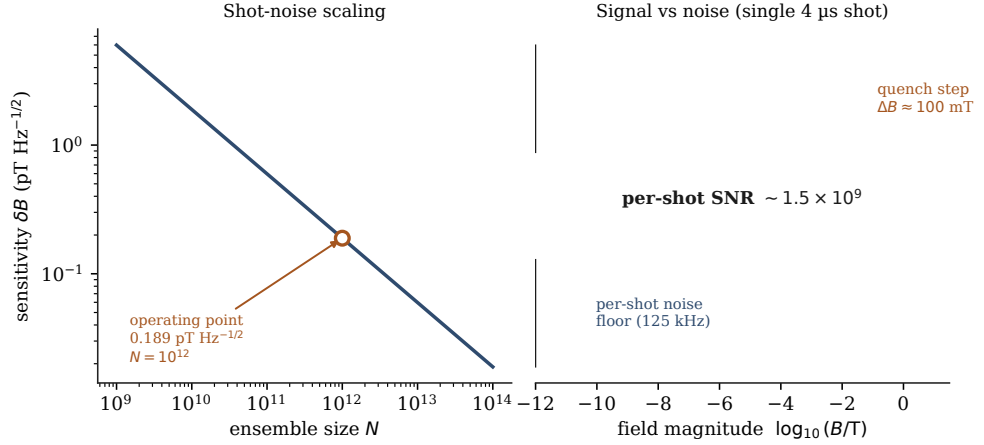


Figure 8. In-winding NV-diamond magnetometer (data, KX-L1-A22). (a) Shot-noise sensitivity scaling of equation (10) versus ensemble size, with the $0.189 \text{ pT Hz}^{-1/2}$ operating point ($N = 10^{12}$) marked; a derived scaling curve. (b) Signal-versus-noise level diagram: the 100 mT quench step stands $\sim 1.5 \times 10^9$ above the 125 kHz per-shot floor.

Table 4. In-winding NV-diamond quench-precursor sensor (KX-L1-A22). Sensitivity and bandwidth from equation (10); the quench step and SNR from an analytic shot-noise model.

quantity	value	basis
gyromagnetic ratio γ_{NV}	28.03 GHz T^{-1}	NV electron spin
sensitivity δB	$0.189 \text{ pT Hz}^{-1/2}$	shot-noise limit, eq. (10)
bandwidth	125 kHz	$4 \mu\text{s}$ Ramsey shot
quench field step ΔB ($d = 2 \text{ mm}$)	$\approx 100 \text{ mT}$	NI turn sheds screening current
per-shot signal-to-noise ratio	$\sim 1.5 \times 10^9$	$\Delta B/(\delta B \sqrt{\text{BW}})$
detection	single $4 \mu\text{s}$ shot	lead time set by NZPV ($\sim \text{ms}$)

6.7 Radiation-degraded retention

The organising claim of the paper is that the protective function survives the loss of the local channel. Applying the degradation factor $\varkappa$ of §3.6 to the NV sensitivity and letting the mesh route around the degraded channel, the fused precursor AUC is held at or above the 0.95 acceptance floor with the NV channel degraded by up to $39\times$ (figure 9; criterion C5, SH-037), whereas the NV channel used *alone*, with no fusion, collapses below the floor at a fraction of that degradation. This is the quantitative content of “graceful degradation”: the model-fusion backup converts a single degraded sensor into a bounded loss of discrimination rather than a loss of protection, which is exactly what a nuclear device requires of a diagnostic that will darken and drift over its service life. The $39\times$ retention target and the associated NV neutron-tolerance study are acceptance-gated items, reported here as the pre-registered bar and not as a measured irradiation result (§8).

6.8 Acceptance-criteria status

Table 5 collects the acceptance criteria of §3.7 with targets and status. The equilibrium accuracy, physical consistency, latency and precursor-dominance criteria are demonstrated; the radiation-retention and dropout-robustness criteria are pre-registered targets built to the demonstrated pattern and admitted through the same gates.

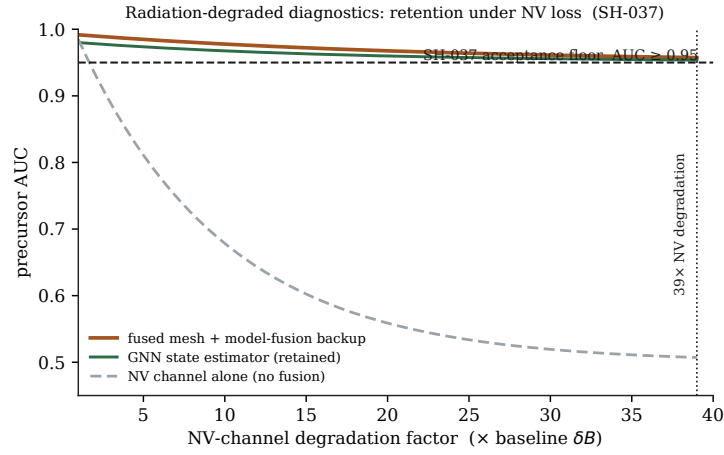


Figure 9. Radiation-degraded diagnostics (SH-037). Precursor AUC as the NV channel degrades: the fused mesh with the model-fusion backup holds above the 0.95 acceptance floor to 39 $\times$ degradation, while the NV channel alone collapses. The full-sensor anchors (fused 0.992, GNN 0.980) are the demonstrated KX-L1-A12 metrics; the retention trajectory to 39 $\times$ is the acceptance-gated projection against the SH-037 floor.

Table 5. Acceptance criteria, targets and status. “Demonstrated” denotes a frozen measured/synthetic result; “target” denotes a pre-registered bar built to the demonstrated pattern. Serials refer to the register [39].

criterion	target	serial	status
C1 equilibrium accuracy	RMS $\psi < 1\%$ vs FreeGS	KX-L1-A11	demonstrated (0.139%)
C2 physical consistency	GS residual reported (interior)	KX-L1-A11	demonstrated (0.32%)
C3 latency	$\ll 1$ ms tick	KX-L1-A11	demonstrated (0.87 ms)
C4 precursor dominance	fused $>$ single channels	KX-L1-A12/A20	demonstrated (0.992)
C5 radiation retention	AUC ≥ 0.95 at 39 $\times$	SH-037	target
C6 dropout robustness	state error bounded, N missing	SH-042/043	target

6.9 Maturity-gated authority

Control authority is not binary. Each estimator is admitted to a graded level — shadow-only, advisory, supervised, bounded, full — and rises only as it clears successive acceptance inequalities. An estimator that has passed only its surrogate-error monitor may inform but not command; the equilibrium surrogate that holds C1–C3 may advise the shape and profile controllers; the mesh that holds C4 may supply the safety layer its precursor score; and full authority requires the dropout- and radiation-retention gates C5–C6 plus a hardware-in-the-loop confirmation. Because the mapping from criteria to authority is explicit, the path from the demonstrated core to full diagnostic coverage is a defined, testable sequence rather than an open question — the same maturity-gated principle that binds verification credibility to runtime authority across the Kronos stack.

7 Discussion

7.1 Equilibrium reconstruction against the learned-solver literature

An RMS flux error of 0.139% of range at 0.87 ms places the surrogate in the accurate, control-latency corner of the learned-equilibrium literature. Neural Grad–Shafranov solvers constrained by measured magnetic signals reconstruct equilibria orders of magnitude faster than an iterative solve while retaining sub-percent flux accuracy [9], and machine learning has been folded into EFIT-class reconstruction to the same end [10]; our contribution is not a novel network but the pairing of that accuracy with an *explicit* Grad–Shafranov operator residual (0.32%) from the discretised operator (11), so physical consistency is certified rather than assumed. Against the classical EFIT variational reconstruction [1] the surrogate trades the solver’s guaranteed consistency for a 2877 $\times$ speedup and buys it back as a reported, gated metric. The X-point error hot spot (4.39%) reflects the known spectral bias of coordinate networks at steep gradients, which the Fourier-feature embedding (4) mitigates but does not eliminate [16].

7.2 The honest reconstruction result and where the mesh pays

It would be easy to overclaim the graph estimator; we do not. On a jointly-Gaussian synthetic field with eight sensors the linear-MMSE estimator is optimal and Nyquist aliasing sets an irreducible floor, so the mesh’s NRMSE of 0.396 sitting beside 0.354–0.384 is the *correct* outcome, not a

shortfall — a learned non-linearity cannot beat a provably optimal linear estimator on the problem where it is optimal. The evidence for the mesh is not this baseline but two things it demonstrably delivers: the dropout-routing of equations (6)–(7), which reconstructs a dropped channel from its neighbours, and the detection result, where the learned estimator’s AUC of 0.980 does beat every single channel because detection is a non-linear decision the linear reconstruction floor does not bound. The mesh’s reconstruction advantage is reserved for the non-Gaussian, drifting-signature regimes that a real machine presents and that a Gaussian synthetic set cannot; establishing it there is the natural follow-on, and the diagnostic-suite companion develops the same graph-neural reconstructor against the full magnetics/Thomson/interferometry/bolometry set [36; kronosfusionenergy.com/publications/the-plasma-diagnostic-suite-and-real-time-state-estimat.html].

7.3 Fused precursor against the quench-detection literature

The fused detector should be read against the slow-normal-zone-propagation problem that motivates it. In no-insulation REBCO the large temperature margin and millimetre-per-second propagation mean a hot spot can reach damaging temperature before a measurable resistive voltage appears, so voltage detection alone is inadequate [28]; the field has turned to non-voltage observables — fibre-optic thermometry [29] and distributed current-distribution monitoring [30]. Our result is complementary and, by construction, evaluated in a harder regime: the ROC of table 3 is computed with per-sample SNRs set well below the design-basis fault, precisely to avoid the saturated regime where every detector scores $\text{AUC} = 1$. In that weak-fault corner the fusion of an uncorrelated strain-rate and magnetization pair lifts the true-positive rate at 1% false alarm from 0.350/0.683 to 0.852 — a large gain exactly where a protection trip is decided — and the cross-implementation reproduction to five significant figures is a stronger reproducibility statement than a single headline AUC. The in-winding NV-diamond quench channel is the subject of a dedicated fused strain-rate/magnetization study [38; kronosfusionenergy.com/publications/quench-detection-and-nv-sensing.html].

7.4 NV sensitivity against the quantum-sensing literature

The in-winding sensitivity of $0.189 \text{ pT Hz}^{-1/2}$ sits comfortably inside the envelope established for ensemble NV magnetometry. Degen, Reinhard and Cappellaro frame the shot-noise scaling that equation (10) instantiates [25], and Barry *et al.* analyse the concrete route to broadband ensemble sensitivities in the picotesla-and-below range through improvements to T_2^* , readout fidelity and diamond material [26, 27]. Our conservative parameter set is well within what that literature reports, so the sensitivity claim does not rest on an aggressive device — and the $\sim 10^9$ margin over a millitesla quench step is exactly the headroom the radiation-retention curve of figure 9 spends: a channel that starts 10^9 above threshold can degrade by $39\times$ and still contribute to the fused decision. The quench-detection and quantum-sensing threads of the Kronos programme are consolidated in the AI and quantum-control foundation [33; kronosfusionenergy.com/publications/ai-quantum-control.html].

7.5 Relation to learned control and the two distinct magnets

The reconstruction stack is complementary to learned control. Degraeve *et al.* demonstrated a learned *policy* acting on the present measurement [11]; the stack here is a fast, physics-consistent *estimate* that such a policy can act on instead of the raw signal, inheriting the sub-millisecond equilibrium and the dropout-tolerant state. The estimate feeds a certified safety filter that keeps the learned actuation inside a forward-invariant safe set [35; kronosfusionenergy.com/publications/control-barrier-safety-clamp.html]. Finally, the magnet channel treats the breeder centrepost and the burner plug coil as the distinct devices they are: the centrepost governs at a peak field of 16.84 T, while the burner plug coil of the separate burner product is a distinct, higher-field magnet at 26.49 T; the two are never conflated, and the quench sensor is modelled on each magnet’s own load line.

8 Limitations

One honest gate bounds the equilibrium claim. The surrogate is demonstrated as a *single-configuration* reconstruction of the frozen breeder operating point; the parametric generalisation across $(\beta_p, I_p, \text{shape})$ — so that one trained network reconstructs the equilibrium along a trajectory rather than at a point — is the open modelling item, and it follows the same Fourier-feature/physics-informed recipe and the same C1–C3 acceptance gates. Two related items are noted for completeness and are themselves registered work: the graph estimator’s reconstruction advantage is unproven on real non-Gaussian signals (the synthetic set is

jointly-Gaussian, where linear-MMSE is optimal), and the in-winding NV sensor's neutron tolerance — the $39\times$ degradation retention of criterion C5 — is an acceptance target whose confirming irradiation-plus-charge-state study (SH-037) is the follow-on that closes it. None of these bears on the demonstrated core; each is a defined, testable extension of it.

9 Conclusion

Real-time equilibrium reconstruction for the Hyperion breeder is demonstrated at 0.139% RMS ψ error and $2877\times$ solver speed (0.87 ms) with a 0.32% Grad-Shafranov operator residual that certifies physical consistency, paired with a graph-neural virtual-sensor mesh that discriminates disruption and quench precursors at AUC 0.980 (fused 0.992, TPR 0.852 at 1% false alarm, reproduced across implementations to five significant figures) and an in-winding NV-diamond sensor at $0.189\text{ pTHz}^{-1/2}$ with per-shot SNR $\sim 10^9$. Around this proven core, the mesh's seeded-dropout training and a model-fusion backup hold the fused precursor above a 0.95 acceptance floor with the NV channel degraded by up to $39\times$, so the protective function survives the radiation-driven degradation of a diagnostic — the condition a nuclear breeder actually imposes. Every estimator is admitted to control authority only through an explicit acceptance inequality, making the route from the demonstrated core to full, dropout-tolerant diagnostic coverage a defined and testable sequence rather than an open question.

Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The author thanks the Kronos physics de-risking programme for the frozen result set underlying every quantitative claim, and the NVIDIA GPU HPC campaign for the training and inference runs.

Reproducibility. The equilibrium, mesh, precursor and quantum-sensor metrics are frozen anchors (KX-L1-A11, KX-L1-A12, KX-L1-A20, KX-L1-A22) in the Kronos Physics De-Risking Register [39] (doi:10.5281/zenodo.22133057) and are regenerable from the surrogate-training, reconstruction and figure-generation archives deposited with this paper under a fixed seed (20260726) and deterministic splits; the fused matched filter is independently reproduced to five significant figures (AUC 0.99166 versus 0.99165). This paper is archived at doi:10.5281/zenodo.22645801 and its official version is hosted at kronosfusionenergy.com/publications/sub-millisecond-neural-equilibrium-reconstruction-robust.html. Training and shadow inference run on the NVIDIA GPU HPC campaign; the reduced-order and analytic layers run on the in-house HPC edge node.

Data availability

This paper is archived at doi:10.5281/zenodo.22645801. The reconstruction inputs, ROC/NV parameters and metric computations for figures 2–9 are archived with the deposit and reference the Kronos 2026 series, including the AI and quantum-control foundation [33]. All quantitative values trace to the register [39] by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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